

Symmetry-Complement Coordinates Across Altermagnets: Predeclared References and Cross-Material Falsification

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A portable symmetry argument must survive materials in which the microscopic hierarchy, probe physics, and even the proposed magnetic state differ, and it must not be made true by choosing its reference model after seeing the target response. We formulate a hardened cross-material falsification benchmark for altermagnets that separates six experimentally distinguishable propositions: static magnetic order M , identification of a response-specific Hamiltonian/operator source sector S , nonrelativistic electronic spin splitting E , chiral collective dynamics χ , a signed reversal or control relation R , and static multipolar content K . The claim vector $\mathbf{c} = (M, S, E, \chi, R, K)$ is not an evidence score. Source-sector attribution is accepted only relative to a predeclared or independently anchored reference H_0, G_0 for which the target channel is absent or symmetry-degenerate without target-specific retuning. We benchmark MnF_2 , FeF_2 , $\alpha\text{-MnTe}$, CrSb , $\alpha\text{-Fe}_2\text{O}_3$, and bulk RuO_2 , and add chiral MnSi as a non-altermagnetic specificity control. FeF_2 is adversarial because its dominant visible magnon splitting is dipolar while the altermagnetic contribution is much smaller; MnSi is adversarial because a strong chiral neutron response arises from Dzyaloshinskii–Moriya physics without an altermagnetic source. The comparison supports a restricted source-sector principle while rejecting a universal one-dimensional coordinate, a universal smallness ratio, and the inference $\chi \neq 0 \Rightarrow$ altermagnetic source. The hardened protocol is: fix the material state and reference independently, identify the source sector, state the measurement map, test the dual counterfactual, challenge mechanism specificity, and require independent closure. Symmetry determines what is allowed; a predeclared adversarial benchmark determines what has actually been identified.

I. INTRODUCTION

Altermagnetism has rapidly developed from a symmetry classification into an experimentally diverse program spanning static magnetic order, momentum-dependent electronic spin splitting, anomalous transport, chiral collective excitations, domain control, and higher-rank multipolar structure.[1, 2] Those observables are related by a common broken-symmetry state, but they are not measurements of the same tensor and they do not share identical nuisance mechanisms. This distinction has become particularly important as several canonical materials now possess both strong positive signatures and sharply constraining null results.

A recurring theoretical idea is that an observable can be controlled by a Hamiltonian term that is small in the total energy yet leading in a selected symmetry channel. In its cleanest form, one introduces a reference Hamiltonian H_0 with an enhanced symmetry G_0 , decomposes the operator space into invariant and non-invariant sectors, and asks which coefficient in the symmetry complement is the first allowed source of a target response. This logic is useful in MnF_2 , where a small seventh-neighbor exchange imbalance has emerged as a candidate source coordinate for an altermagnetic chiral response within a specified spin-wave model.[3, 4] The key question for the present work is whether that construction survives outside MnF_2 without being promoted into an unjustified universal numerical rule.

The answer requires a falsification benchmark rather than a collection of favorable examples. The benchmark must allow at least four outcomes: a one-dimensional source coordinate may survive; only a higher-dimensional source sector may survive; the target observable may be established while its microscopic source remains model-dependent; or the proposed material state itself may fail before downstream observables can be attributed to it. The last case is essential. A general framework that cannot stop at an upstream contradiction is not genuinely falsifiable.

We therefore extend a previous claim-specific closure architecture by making the *source sector* explicit. A candidate material is represented by

$$\mathbf{c} = (M, S, E, \chi, R, K), \quad (1)$$

where M denotes the static magnetic state under study; S denotes identification of a Hamiltonian/operator sector whose symmetry character is necessary for a target response; E denotes a nonrelativistic momentum-dependent electronic spin-splitting claim; χ denotes chiral or symmetry-split collective dynamics; R denotes a signed reversal or controlled transformation; and K denotes static multipolar content or rank. The six components are deliberately not collapsed into a scalar. An open K does not erase a closed E ; a transport anomaly does not repair a contradicted M ; and an experimentally visible χ does not by itself prove that a unique microscopic coefficient has been isolated.

We apply this logic to six candidate altermagnetic systems spanning insulating, semiconducting, metallic, positive, partial, and negative cases: MnF_2 , FeF_2 , $\alpha\text{-MnTe}$,

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CrSb, hematite, and bulk RuO₂. MnF₂ tests whether a sharp source-coordinate construction survives a probe-specific spectral null. FeF₂ is deliberately adversarial: very-high-resolution INS shows that long-range dipolar interactions dominate the visible magnon splitting, while a much smaller altermagnetic contribution emerges only where the dipolar splitting vanishes.[5] MnTe tests closure across bulk order, electronic structure, directly observed chiral magnons, and active domain control.[6] CrSb tests a metallic g -wave system with independent ARPES, RIXS, neutron, and real-space multipolar probes. Hematite tests a partial case in which a magnon source family can be modeled without closing the full electronic or static-multipole program. RuO₂ supplies the upstream negative control: multiple bulk probes favor a paramagnetic state, while strained films remain strongly preparation-dependent.[7–10]

We also add MnSi as a specificity control outside the altermagnet claim matrix. Polarized-neutron and inelastic-neutron measurements establish chiral magnetic response and nonreciprocal magnons in a noncentrosymmetric chiral magnet whose microscopic origin is Dzyaloshinskii–Moriya interaction.[11, 12] Therefore a successful benchmark must explicitly reject the implication $\chi \neq 0 \Rightarrow S_{\text{alt}} \neq 0$.

The resulting benchmark does not establish a universal hiddenness ratio, a universal inverse amplification law, or a universal single coordinate λ . Instead it tests the more conservative statement that target responses can often be organized by symmetry-complement *sectors*, provided the reference is fixed independently of the target and competing mechanisms are challenged explicitly.

II. SOURCE-SECTOR FORMALISM AND FALSIFICATION LOGIC

A. Predeclared-reference admissibility

A symmetry complement is meaningful only relative to a stated reference. If H_0 is chosen after inspecting the target response so that the successful interaction is, by construction, the “complement,” the source principle becomes nearly tautological. We therefore require a prospective source claim to fix $(H_{0,m}, G_{0,m}, \mathcal{V}_m)$ before fitting the target channel. For a retrospective literature audit, the weaker but testable requirement is that the reference be *independently anchored*: by crystallographic symmetry, an independently justified structural limit, a Hamiltonian fitted to orthogonal observables, or a controlled symmetry-restoring operation.

For target O_j define a Boolean reference-admissibility certificate

$$\mathfrak{A}_{m,j} = A_{\text{ind}} \wedge A_{\text{state}} \wedge A_{\text{null}} \wedge A_{\text{no retune}}, \quad (2)$$

where A_{ind} requires target-independent motivation of the reference; A_{state} requires the same declared material state on both sides of the counterfactual; A_{null} requires O_j to

vanish or become symmetry-degenerate in the reference; and $A_{\text{no retune}}$ forbids compensating target-specific refits of unrelated parameters after the candidate source is removed. If $\mathfrak{A}_{m,j} = 0$, the source-sector claim cannot receive a CLOSED or PASS verdict in this benchmark.

This rule does not demand that a historical paper literally preregister its Hamiltonian. It demands that the reference used for inference could have been specified without knowledge of the target outcome. The distinction is operational: a high-symmetry exchange model fixed by independent spectra can be admissible; a reference obtained by deleting whichever term happens to reproduce the desired anomaly is not.

B. From a coordinate to a sector

For material m , choose a reference Hamiltonian $H_{0,m}$ with reference symmetry group $G_{0,m}$ and operator space $\mathcal{V}_m$. Group averaging defines a projector onto the invariant subspace,

$$\mathcal{P}_m[X] = \frac{1}{|G_{0,m}|} \sum_{g \in G_{0,m}} U_g X U_g^{-1}, \quad (3)$$

with the decomposition

$$\mathcal{V}_m = \text{im } \mathcal{P}_m \oplus \ker \mathcal{P}_m. \quad (4)$$

The symmetry-complement source sector is the physically relevant part of $\ker \mathcal{P}_m$ for a specified target response. Write

$$H_m = H_{0,m} + \sum_{a=1}^{d_m} \lambda_{m,a} V_{m,a} + H_{m,\text{other}}. \quad (5)$$

A single scalar source coordinate is therefore the special case $d_m = 1$. Cross-material generalization should not assume $d_m = 1$, nor that $|\lambda_{m,a}|$ is small relative to every dominant coupling.

For an observable O_j define the local source sensitivity vector

$$(g_{m,j})_a = \left. \frac{\partial \langle O_j \rangle}{\partial \lambda_{m,a}} \right|_{\lambda=0}. \quad (6)$$

If symmetry forces $O_j = 0$ at the reference point, then a nonzero $\mathbf{g}_{m,j}$ identifies a first-order direction in source space. For a symmetry operation that maps $\boldsymbol{\lambda} \rightarrow -\boldsymbol{\lambda}$ and $O_j \rightarrow -O_j$, analyticity gives an odd expansion along that direction,

$$\langle O_j \rangle_{t\hat{\lambda}} = \chi_1 t + \chi_3 t^3 + \dots \quad (7)$$

This is a selection statement, not an amplification law. A small t can dominate a selected channel because lower-order invariant contributions are forbidden, not because the microscopic coefficient has become large.

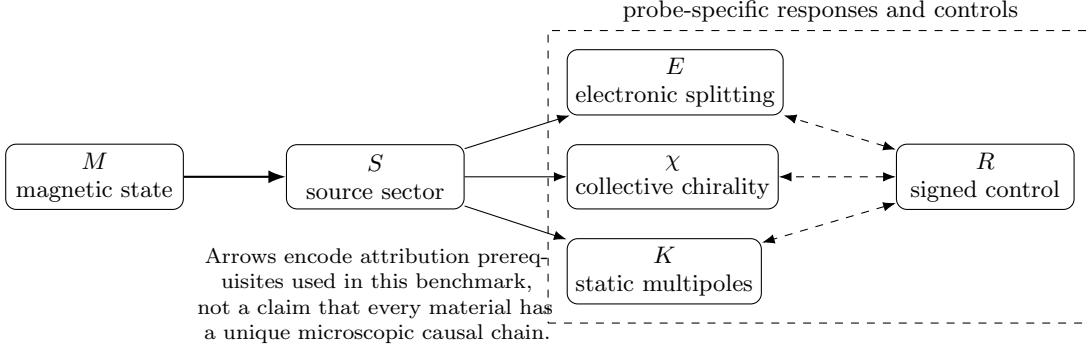


FIG. 1. Six-component closure architecture. M and S are upstream attribution questions: does the stated material state exist, and has a symmetry-resolving Hamiltonian/operator sector been identified for the response under discussion? E , χ , and K are distinct response or order-parameter propositions, while R records a signed reversal or controlled transformation. A component can remain open without invalidating independently closed components.

C. Operational closure of S

The source-sector component S is treated conservatively. We call it CLOSED only when $\mathfrak{A}_{m,j} = 1$ and experiments plus modeling isolate the relevant source direction or sector with a strong counterfactual: removing or symmetry-restoring that sector eliminates the target response, competing operators are independently constrained, and the result survives at least one signed or forbidden dual. We use SUPPORTED when an independently anchored fitted or first-principles Hamiltonian identifies a plausible source family but uniqueness remains model-dependent. We use BLOCKED when the upstream magnetic state required for the attribution is itself contradicted. A post-hoc reference is insufficient even if it reproduces the data.

This distinction is important because different observables can have different source maps in the same material. A Hamiltonian term controlling magnon chirality need not be the same object that controls the largest electronic band splitting or the leading static multipole. The correct generalization is therefore response-specific:

$$S_{m,j} : \lambda_m \mapsto O_{m,j}, \quad (8)$$

not a universal material scalar S_m with a single number attached.

D. Measurement map and admissible sets

For material state m and probe p , write the recorded data as

$$\mathbf{y}_{mp} = \mathcal{T}_{mp}[\mathcal{P}_{mp} \mathbf{r}_m(\boldsymbol{\theta}_m)] + B_{mp} \boldsymbol{\nu}_{mp} + \boldsymbol{\epsilon}_{mp}, \quad (9)$$

where $\mathbf{r}_m$ denotes physical response tensors, $\mathcal{P}_{mp}$ the probe projection, $\mathcal{T}_{mp}$ the transfer/resolution map, $B_{mp} \boldsymbol{\nu}_{mp}$ structured nuisance contributions, and $\boldsymbol{\epsilon}_{mp}$ statistical error. A null may therefore arise from the physical response, the source coordinate, projection, resolution, domains, or an incorrect material-state hypothesis.

For hypothesis H and tolerance τ_p , define the probe-specific admissible set

$$\mathcal{A}_p(H) = \left\{ \boldsymbol{\theta} \in H : \min_{\boldsymbol{\nu}_p} \chi_p^2(\boldsymbol{\theta}, \boldsymbol{\nu}_p) \leq \tau_p \right\}. \quad (10)$$

Cross-probe closure requires

$$\mathcal{A}_{\text{joint}}(H) = \bigcap_{p \in \mathcal{P}_H} \mathcal{A}_p(H) \neq \emptyset, \quad (11)$$

while predeclared alternatives must fail at least one sufficiently independent constraint. Equation (11) is book-keeping rather than a universal statistical threshold; its purpose is to keep physical assumptions and nuisance directions explicit.

E. Witness–obstruction duality

Every positive witness should be paired with a dual obstruction: a forbidden momentum direction, reversed domain, reversed polarization, temperature regime, nonmagnetic calculation, multiple-scattering test, or symmetry-restored Hamiltonian. A permitted signal without its obstruction is rarely identifying. Conversely, a null in one projected channel does not automatically negate a response visible through another measurement kernel.

This duality becomes especially useful for S . The positive statement “this operator produces the response” should be paired with the counterfactual “the response vanishes or changes character when the operator is removed, symmetrized, or reversed.” Where the counterfactual remains entirely model-assisted, S should remain SUPPORTED rather than CLOSED.

III. BENCHMARK PROTOCOL

The benchmark is a structured literature audit, not a numerical meta-analysis. The literature cutoff is September 12, 2026. Peer-reviewed results are preferred; recent

preprints are retained only when their provisional status is explicit. The status labels are proposition-specific:

- **CLOSED**: direct constraint plus a strong signed control, symmetry null, or independent rejection of a leading alternative;
- **SUPPORTED**: direct evidence or model-backed source identification with material interpretive dependence remaining;
- **PROVISIONAL**: recent preprint-level evidence not yet folded into the mature peer-reviewed set;
- **OPEN**: not directly established in the benchmark literature;
- **CONTRADICTED**: multiple independent current probes reject the stated material-state proposition;
- **DISFAVORED**: current evidence weighs strongly against the proposition without constituting an exhaustive exclusion;
- **BLOCKED**: upstream prerequisites needed for the attribution are contradicted;
- **STATE-DEPENDENT**: the result changes materially with strain, orientation, thickness, stoichiometry, substrate, surface, or defect state.

These labels do not rank entire materials. They identify which physical proposition has been constrained.

We separately assign an adversarial benchmark verdict for the source principle. A **PASS** requires an admissible reference, a target-channel null or degeneracy in that reference, a response when the candidate sector is restored, and at least one dual or orthogonal constraint that discriminates the proposed source from a leading alternative. A **PARTIAL** means the same architecture is supported but one decisive step remains model-assisted or non-unique. **BLOCKED** means an upstream state or reference condition fails before source inference is meaningful. A **SPECIFICITY PASS** is reserved for an external control in which the downstream signature is present but an independently known non-altermagnetic source correctly prevents altermagnetic attribution. These verdicts test the framework itself; they are not another evidence score.

IV. MnF_2 : A SHARP SOURCE COORDINATE INSIDE A BOUNDED MODEL

MnF_2 remains the calibration case because the source and readout questions can be separated unusually cleanly. Unpolarized INS reported no resolvable altermagnetic magnon band splitting within the experimental resolution and found that a higher-symmetry exchange model reproduced the observed spectrum.[13] Polarized INS later accessed an antisymmetric dynamical channel and reported a domain-dependent chiral response with opposite signs

on the two magnon modes.[3] These statements are compatible because unresolved positive spectral weights and projection onto an antisymmetric polarization channel are different measurement operations.

The new S component sharpens what can and cannot be inferred from this coexistence. In the range-seven spin-wave analysis associated with the polarized-neutron work, the inequivalence of symmetry-distinct seventh-neighbor exchange paths produces the relevant altermagnetic chiral term. A separate 2026 first-principles total-energy mapping identifies a seventh-neighbor imbalance of order several μeV and proposes signed field-based readouts.[4] Within that bounded Hamiltonian class, the source coordinate can be written schematically as

$$\delta J_7 = J_{7b} - J_{7a}, \quad (12)$$

with sign meaningful only after the bond-label convention is declared. The source claim is strong but not universal: the result does not prove that every higher-order interaction or every enlarged MnF_2 Hamiltonian reduces to this one coordinate. We therefore classify S as **SUPPORTED**, not as a theorem of the full material Hamiltonian.

The static sector remains distinct. Polarized-neutron diffraction now reconstructs an anisotropic Mn magnetization and ligand spin polarization consistent with a symmetry-allowed rank-5 magnetic multipole contribution.[14] This preprint strengthens K while also demonstrating why K , S , and χ must remain separate. A seventh-neighbor exchange coefficient is not a magnetic multipole, and a dynamical chiral correlator is not a direct static-rank measurement.

MnF_2 therefore gives the strongest example of a nearly one-dimensional source coordinate in this benchmark, but it simultaneously warns against universalizing that dimensionality. The hardened reference test is conditional: the equal-path or symmetry-restored J_7 reference must be fixed as part of the bounded Hamiltonian class before the chiral target is used for discrimination. With that condition, the source-sector architecture receives a **PARTIAL** rather than an unconditional theorem-level pass.

V. FeF_2 : ADVERSARIAL SEPARATION OF DIPOLAR AND ALTERMAGNETIC SOURCES

FeF_2 is the most useful neighboring adversary because it shares the rutile antiferromagnetic setting but does not permit the visible magnon splitting to be equated with the altermagnetic source. Very-high-resolution INS finds that the dominant splitting is produced by the long-range dipolar interaction. Only at momenta where the dipolar contribution vanishes is an additional broadening observed and attributed to an altermagnetic chiral splitting of approximately $35 \mu\text{eV}$. Polarized measurements further show that where the dipolar term is active the nominal chiral modes are strongly mixed and predominantly linearly polarized, with at most a small chiral component.[5]

This case forces the reference model to retain an independently established nuisance mechanism rather than deleting it for convenience. An admissible source test therefore has the schematic structure

$$H_{\text{ref}} = H_{\text{exch}}^{(0)} + H_{\text{dip}} + H_{\text{ani}}, \quad H = H_{\text{ref}} + H_{\text{alt}}, \quad (13)$$

where H_{dip} remains present in both hypotheses. The decisive kinematic control is a momentum set $\mathcal{Q}_0$ at which the dipolar splitting is symmetry- or geometry-suppressed,

$$\Delta_{\text{dip}}(\mathbf{Q} \in \mathcal{Q}_0) \simeq 0, \quad \Delta_{\text{res}}(\mathbf{Q} \in \mathcal{Q}_0) > 0. \quad (14)$$

A benchmark that instead assigned the entire large visible splitting to “altermagnetism” would fail this adversarial case.

We therefore classify M as CLOSED, S as SUPPORTED, and χ as SUPPORTED; E , a controlled signed reversal R , and static multipolar rank K remain OPEN in the present audit. FeF_2 receives a PASS for mechanism separation but not coordinate closure: it demonstrates that a source-sector benchmark must discriminate a weak target source from a larger, independently known interaction that acts in the same spectrum.

VI. α -MNTE: MULTICHANNEL CLOSURE WITHOUT A UNIVERSAL WEAK COORDINATE

Hexagonal MnTe supplies a qualitatively different positive case. ARPES directly established large momentum-dependent lifting of Kramers degeneracy associated with the ordered altermagnetic phase, with complementary experiments reporting temperature-dependent spin splitting up to hundreds of meV.[15, 16] These measurements strongly close E .

INS first resolved the split magnon spectrum and showed that the lifted degeneracy is reproduced by symmetric exchange.[17] In 2026, polarized inelastic neutron scattering directly observed the handedness of the split modes and reversibly switched the magnon chirality with magnetic-field control.[6] The collective proposition χ is therefore CLOSED in the present benchmark. This does not reduce the entire MnTe problem to a unique weak scalar analogous to MnF_2 ’s δJ_7 : the relevant microscopic source is better treated as an exchange/symmetry sector of the fitted Hamiltonian. Accordingly, S remains SUPPORTED because no universal one-dimensional coefficient has been isolated across the electronic and magnonic channels.

The 2026 experiments substantially strengthen M and R . Polarized-neutron diffraction directly probed the bulk altermagnetic order and demonstrated field-cooling switching of the Néel-vector domain together with a coupled weak ferromagnetic moment.[18] A separate strain-control study used neutron scattering and transport to detwin the in-plane magnetic domains and tune, including reverse, the anomalous Hall response.[19] The signed-control proposition is therefore not inferred from a transport

anomaly alone; it is tied to an independently measured change of the magnetic domain state.

MnTe falsifies one possible overgeneralization of the MnF_2 lesson. A material can satisfy the source-sector principle while lacking a distinguished, universally small λ . The portable object is the symmetry character of the source sector and its response map, not the numerical ratio of one coupling to the dominant exchange.

VII. CRSB: ORTHOGONAL CLOSURE IN A METALLIC g -WAVE SYSTEM

CrSb provides the broadest independent-nuisance closure in the present benchmark. Soft-x-ray ARPES directly observes a large momentum-dependent electronic band splitting, of order tenths of an eV, along a low-symmetry path while symmetry-related high-symmetry directions remain degenerate.[20] Thin-film work subsequently showed that the characteristic three-dimensional band splitting survives to nanometer-scale epitaxial films without a net magnetization.[21]

The collective sector is independently constrained. Circularly polarized RIXS maps a pronounced magnon circular dichroism whose sign changes with momentum and antiferromagnetic domain and which vanishes in symmetry-required directions.[22] The branch energy separation can remain below the instrumental linewidth while the polarization channel retains strong contrast, providing a direct cross-material example of projection and resolution acting on different observables. INS has also reported low-symmetry chiral splitting, and a 2026 analytical spin-wave treatment relates the measured spectrum to a compact interaction model.[23, 24]

For S , the safest conclusion is again a sector rather than a universal coordinate. The g -wave symmetry requires higher-order exchange structure beyond the minimal interactions that reproduce only high-symmetry dispersions, while current models identify interaction combinations capable of producing the low-symmetry splitting. The source is therefore SUPPORTED, but uniqueness is not yet experimentally closed.

CrSb also hosts the newest direct static-multipole evidence. A September 2026 preprint combines high-energy synchrotron x-ray diffraction, valence-electron-density analysis, and polarized-neutron diffraction to visualize a g -wave multipolar pattern and connect it microscopically to the electronic spin splitting.[25] Because the result is recent, K remains PROVISIONAL in the benchmark. Taken together, CrSb shows the value of orthogonal probes: ARPES, RIXS, INS, x-ray density reconstruction, and polarized neutrons do not share one dominant nuisance model.

VIII. HEMATITE: A RESPONSE-SPECIFIC SOURCE FAMILY

Hematite (α -Fe₂O₃) is included precisely because it is incomplete. INS resolved an approximately 3 meV magnon splitting near 100 meV; first-principles calculations and spin-wave modeling traced the effect to alternating thirteenth-neighbor exchange interactions and assigned opposite chirality to the split branches.[26, 27] Later first-principles and atomistic-dynamics work further developed the magnonic description.[28]

For the present benchmark this gives a useful pattern: M is established, χ is SUPPORTED, and S is SUPPORTED for the magnon response because a concrete alternating-exchange source family is identified. But E , R , and K remain open in the sense used here. It would be incorrect to promote the magnon source family into evidence for an independently unmeasured electronic splitting or a unique static multipole rank.

Hematite therefore tests response-specificity. The source-sector principle survives, but closure of $S_{m,\chi}$ does not automatically close $S_{m,E}$ or $S_{m,K}$. That locality is a feature of the framework rather than a weakness.

IX. RUO₂: THE UPSTREAM NEGATIVE CONTROL

Bulk stoichiometric RuO₂ is the most important negative control because it was historically promoted as a prototypical metallic altermagnet while its magnetic ground state has come under increasingly stringent scrutiny. Keßler *et al.* combined μ SR, neutron diffraction, and first-principles analysis and reported no long-range magnetic order, placing very small upper bounds on the ordered Ru moment and identifying multiple scattering as the origin of an earlier neutron feature.[7] High-quality bulk transport subsequently found no magnetic transition below 400 K, no crystal Hall effect, and substantially better agreement with nonmagnetic than altermagnetic band calculations.[8]

Independent 2026 bulk measurements reinforce that conclusion. Torque magnetometry and magnetization are inconsistent with a collinear Néel state along either the c axis or the basal plane and reveal quantum oscillations compatible with a paramagnetic electronic structure.[9] Angle-dependent quantum oscillations on ultraclean crystals with very large residual-resistivity ratios map a Fermi surface that agrees closely with paramagnetic DFT; prominent high-frequency branches are absent from the corresponding altermagnetic calculation.[10] Because these probes have different nuisance structures, their agreement is more constraining than repetition of one measurement class.

For the proposition “bulk stoichiometric RuO₂ realizes the proposed compensated altermagnetic ground state,” we therefore label M CONTRADICTED. This has a direct logical consequence for S : the altermagnetic source-sector

attribution is BLOCKED until the relevant magnetic state is re-established in the same bulk state. Downstream electronic or transport anomalies cannot compensate for that upstream failure. The proposed bulk altermagnetic electronic splitting is therefore labeled DISFAVORED rather than counted as an independent positive component.

This conclusion must not be exported indiscriminately to thin films. RuO₂ films display strong preparation dependence. Single-variant epitaxial films have been reported with structural, XMLD, and spin-transport signatures interpreted in altermagnetic language.[29] A 2026 Nano Letters study and a 2026 PRB study find that selected compressive-strain/orientation conditions can stabilize magnetic or altermagnetic-like electronic structures.[30, 31] Conversely, another peer-reviewed 2026 first-principles study finds film magnetism fragile and fails to stabilize a compensated antiferromagnetic ground state in the structures it examined.[32] The correct benchmark conclusion is therefore state dependence: every film claim must establish M and S in the same thickness, strain, stoichiometry, orientation, substrate, and relaxation state in which the downstream response is measured.

This is the decisive negative result of Paper V. The framework is allowed to stop at the source-state gate.

X. MNSI: NON-ALTERMAGNETIC CHIRALITY AS A SPECIFICITY CONTROL

A framework that identifies every chiral neutron response as altermagnetic would be sensitive but not specific. MnSi provides the required external control. Its noncentrosymmetric crystal structure supports an antisymmetric Dzyaloshinskii–Moriya interaction, and polarized-neutron measurements directly access an antisymmetric dynamical susceptibility with chiral response.[11] Inelastic neutron scattering also maps strongly nonreciprocal magnons whose field and wave-vector asymmetries are quantitatively described by the low-energy theory of cubic chiral magnets with Dzyaloshinskii–Moriya interaction.[12]

The control therefore realizes

$$\begin{aligned} \chi &\neq 0, \\ S_{\text{DMI}} &\neq 0, \\ S_{\text{alt}} &= 0 \quad \text{for the tested attribution.} \end{aligned} \tag{15}$$

so that

$$\boxed{\chi \neq 0 \not\Rightarrow S_{\text{alt}} \neq 0.} \tag{16}$$

MnSi receives a SPECIFICITY PASS: the benchmark must route the observed chirality to the independently anchored Dzyaloshinskii–Moriya source instead of treating chirality itself as a sufficient witness of altermagnetism. This control places a hard boundary on the interpretation of polarized-neutron and nonreciprocal-magnon signatures throughout the candidate-material set.

TABLE I. Claim-specific closure status for candidate altermagnetic systems at the September 12, 2026 literature cutoff. The matrix is not a scalar ranking. MnSi is treated separately as a non-altermagnetic specificity control. “Blocked” means that an upstream prerequisite for the stated altermagnetic attribution is contradicted; “disfavored” means strong current evidence weighs against the proposition without asserting exhaustive impossibility.

Material	M	S	E	χ	R	K
MnF ₂	CLOSED	SUPPORTED	OPEN	CLOSED	CLOSED	PROVISIONAL
FeF ₂	CLOSED	SUPPORTED	OPEN	SUPPORTED	OPEN	OPEN
α -MnTe	CLOSED	SUPPORTED	CLOSED	CLOSED	CLOSED	OPEN
CrSb	CLOSED	SUPPORTED	CLOSED	CLOSED	CLOSED	PROVISIONAL
α -Fe ₂ O ₃	CLOSED	SUPPORTED	OPEN	SUPPORTED	OPEN	OPEN
RuO ₂ (bulk)	CONTRADICTED	BLOCKED	DISFAVORED	OPEN	STATE-DEPENDENT ^a	OPEN

^aSigned or symmetry-selective responses reported in RuO₂ are strongly material-state dependent; thin-film results are not inherited as evidence for the contradicted bulk magnetic state.

A. Adversarial verdicts

The source-sector principle is additionally judged against explicit hostile counterfactuals. These verdicts test the inference architecture rather than ranking materials:

- **MnF₂** — PARTIAL. The bounded symmetry-restored/equal-path exchange reference is admissible only if fixed independently of the polarized chiral target; the decisive counterfactual sets the seventh-neighbor imbalance to zero without target-specific refitting.
- **FeF₂** — PASS. The reference retains the independently established long-range dipolar interaction, and the target altermagnetic contribution is sought where the dipolar splitting vanishes.
- **α -MnTe** — PASS for χ and R ; S remains partial. The ordered state and exchange model are constrained independently, while polarized INS directly measures and reverses the magnon chirality.
- **CrSb** — PARTIAL. A minimal high-symmetry dispersion model is fixed before the low-symmetry test; the low-symmetry dichroism/splitting requires additional g -wave source structure, but microscopic uniqueness remains open.
- **α -Fe₂O₃** — PARTIAL. The alternating long-range exchange family is tested by model counterfactuals, but direct signed closure remains incomplete.
- **Bulk RuO₂** — BLOCKED. The upstream magnetic-state proposition fails across independent bulk probes before a bulk altermagnetic source sector can be inferred.
- **MnSi control** — SPECIFICITY PASS. Chiral neutron response exists, but the independently anchored Dzyaloshinskii–Moriya source correctly prevents altermagnetic attribution.

XI. CROSS-MATERIAL RESULTS

A. The source principle survives; the universal coordinate does not

Table I exposes the main result. The operator-space idea generalizes most robustly at the level of a *source sector*: in each positive dynamical case, the target response can be associated with interactions or order-parameter structures that are symmetry-distinct from a reference model in which the target channel is absent or degenerate. What does not survive is the stronger hypothesis that every altermagnet possesses a unique, weak, one-dimensional coordinate analogous to MnF₂’s δJ_7 .

The cross-material source structure is schematic but instructive:

$$\begin{aligned}
 \text{MnF}_2 : d_S &\simeq 1, & \text{FeF}_2 : d_S &\geq 1, \\
 \text{MnTe} : d_S &> 1, & \text{CrSb} : d_S &> 1, \\
 \text{Fe}_2\text{O}_3 : d_S &\gtrsim 1, & \text{RuO}_2 : S &\text{blocked}.
 \end{aligned} \tag{17}$$

The symbols in Eq. (17) are organizational, not fitted dimensions. MnF₂ is the bounded range-seven near-scalar case; FeF₂ contains a weak altermagnetic sector behind a larger dipolar contribution; MnTe and CrSb require broader exchange sectors; hematite uses an alternating long-range-exchange family; and bulk RuO₂ is blocked upstream. MnSi is excluded from this dimensionality list because the relevant source is Dzyaloshinskii–Moriya interaction rather than an altermagnetic symmetry-complement sector.

B. No universal evidence scalar and no universal hierarchy ratio

The benchmark gives no justification for a total evidence score

$$S = \sum_i w_i c_i. \tag{18}$$

The weights would be arbitrary and would conflate independent physical propositions. More importantly, an open multipole rank cannot subtract from directly observed electronic splitting, and a strong transport response cannot compensate for a contradicted source state.

For the same reason, no universal hierarchy ratio of the form $|\lambda|/|J|$ emerges. MnF_2 contains a particularly small exchange imbalance; MnTe and CrSb display large observable splittings tied to broader exchange and symmetry structures; hematite requires very long-range exchange in the fitted model. The portable result is therefore not “small interaction $\Rightarrow$ hidden response,” but

$$\boxed{\text{selected response} \Rightarrow \text{source sector} + \text{probe map.}} \quad (19)$$

C. Specificity is not sensitivity

FeF_2 and MnSi expose two complementary failure modes. FeF_2 shows that a large spectral effect can be dominated by an interaction other than the target altermagnetic source; MnSi shows that even a genuine chiral magnetic response can arise from a different microscopic symmetry mechanism. A source benchmark must therefore reject both substitutions,

$$\Delta_{\text{magnon}} \neq 0 \not\Rightarrow S_{\text{alt}} \neq 0, \quad \chi \neq 0 \not\Rightarrow S_{\text{alt}} \neq 0. \quad (20)$$

The appropriate positive claim is conditional: after the material state and reference are fixed independently, the target response must survive source-specific controls and leading alternative mechanisms must fail their own predicted dual tests. This is why the benchmark records S separately from χ and why a positive downstream signature cannot by itself close a microscopic source attribution.

D. Independent nuisance structures are a form of leverage

Cross-probe agreement is strongest when the leading false-positive mechanisms differ. ARPES matrix elements, neutron multiple scattering, RIXS polarization leakage, transport geometry, quantum-oscillation orbit assignment, and real-space density reconstruction do not share identical nuisance subspaces. This does not make any probe infallible, but it makes a common latent state more constrained.

Conversely, nominally different measurements should not be counted as fully independent when they inherit the same sample preparation, strain state, surface reconstruction, domain distribution, or model-dependent band assignment. The benchmark therefore values *orthogonality of nuisance mechanisms*, not simply the number of reported observables.

E. Source-state gating

RuO_2 motivates an explicit attribution rule. Let $W_{m,j}$ denote a witness for response proposition j in material state m , and let $N_{m,j}$ denote unresolved nuisance alternatives. A conservative material-level attribution requires

$$C_{m,j} = M_m \wedge W_{m,j} \wedge \neg N_{m,j}, \quad (21)$$

with $S_{m,j}$ additionally required when the claim concerns a specific microscopic source. Equation (21) is logical bookkeeping, not a probabilistic score. Its practical content is simple: a downstream signal can remain physically real while its attribution to an altermagnetic bulk state fails because the upstream state is absent.

XII. DECISIVE MISSING EXPERIMENTS

The benchmark generates concrete next tests without assuming a universal material metric.

For MnF_2 , the decisive source-side experiment is a signed field or partner-momentum measurement that extracts the sign and scale of the seventh-neighbor imbalance independently of the polarization-odd neutron analysis. Agreement would move S closer to closure; disagreement would localize the failure to the source model rather than the chiral measurement itself.[4]

For FeF_2 , the decisive next step is a source-specific signed control at momenta selected to suppress the dipolar contribution, combined with sufficient resolution to distinguish genuine two-branch structure from broadening. The purpose is not merely to see a small residual feature but to test whether its field, momentum, and polarization transformations follow the proposed altermagnetic source while a dipolar-only model fails.

For MnTe , a polarization-resolved magnon measurement would convert the current model-assisted handedness assignment into a direct χ closure. A real-space multipole reconstruction would independently test K and determine whether the static tensor hierarchy maps cleanly onto the already closed electronic and domain-control propositions.

For CrSb , the most important near-term test is peer-reviewed replication of the new real-space g -wave multipole result and, ideally, a same-crystal comparison with the RIXS or neutron domain map. The source-sector question would be strengthened by a controlled Hamiltonian perturbation, for example strain, that changes the predicted low-symmetry magnon splitting while leaving high-symmetry constraints intact.

For hematite, polarization- or domain-resolved magnon handedness would turn the current model-assisted chirality assignment into a direct signed test. Independent electronic spectroscopy could then determine whether the same material state supports the corresponding nonrelativistic electronic proposition.

For RuO_2 , every thin-film claim should begin upstream: establish the static magnetic source in the *same* sample

state used for transport or spectroscopy. Strain, thickness, stoichiometry, orientation, substrate, and relaxation must be treated as coordinates of the hypothesis, not as incidental sample metadata.

All of these tests share one design principle: measure the witness and its dual obstruction on the same controlled material state.

XIII. DISCUSSION

The present benchmark is deliberately conservative. It does not redefine altermagnetism and does not require every material to exhibit every available signature. Its purpose is to prevent claim migration across distinct tensors and across distinct levels of the source-to-readout chain.

Four lessons follow. First, the reference model must be independently anchored; otherwise a symmetry complement can be manufactured retrospectively. Second, a material can be strongly established in one sector while remaining open in another. MnTe can have closed electronic and domain-control propositions while static multipole rank remains open. Hematite can have a well-supported magnon source family while its electronic proposition remains untested in this benchmark. Third, a null is local to its measurement map. MnF₂ and CrSb both illustrate that insufficient spectral resolution need not remove a polarization-selective response channel, while FeF₂ shows that a large visible splitting can belong mainly to a competing interaction. Fourth, an upstream contradiction has a different logical status from a downstream projection null. Bulk RuO₂ is not merely “another null experiment”; its magnetic-source proposition is disfavored by several orthogonal bulk probes, so downstream altermagnetic attribution is blocked until that upstream condition is repaired.

The source-sector addition also clarifies what the operator-space principle can legitimately claim. A symmetry complement is a decomposition of operator space relative to a chosen reference symmetry; because that factorization is reference-dependent, the reference must be predeclared prospectively or independently anchored retrospectively. The construction does not guarantee a universal material-independent basis, a one-dimensional complement, or a hierarchy in which all relevant coefficients are small. The most defensible cross-material statement is therefore structural:

$$H_0, G_0 \implies \mathcal{V} = \mathcal{V}_{\parallel} \oplus \mathcal{V}_{\perp}, \quad (22)$$

followed by a material- and observable-specific question: which directions in $\mathcal{V}_{\perp}$ have nonzero response derivatives for the target observable, and which experiments distinguish those directions from nuisance perturbations?

This is also why the claim vector should remain a lattice rather than a score. Each component can be updated independently as new experiments arrive. A future bulk magnetic signal in RuO₂ that survives μ SR, neutron,

torque, transport, and Fermi-surface alternatives would reopen M and thereby unblock S ; a failed replication of the CrSb multipole preprint would weaken K without erasing the ARPES or RIXS results. Locality of revision is a feature of a falsifiable framework.

XIV. CONCLUSION

A cross-material test of symmetry-selective observability should be allowed to fail, including by failing its own choice of reference. We therefore hardened the source-sector benchmark with a predeclared-reference requirement, mechanism-specific counterfactuals, and an external chirality control, while retaining six independent propositions,

$$\mathbf{c} = (M, S, E, \chi, R, K). \quad (23)$$

The candidate-material set now includes MnF₂, FeF₂, MnTe, CrSb, hematite, and bulk RuO₂, with MnSi serving as a non-altermagnetic specificity control.

The restricted generalization survives. Response-specific symmetry-complement source sectors remain useful organizing objects, but a universal weak one-dimensional coordinate is not supported. MnF₂ is unusually close to that special case within a bounded spin-wave model; FeF₂ demonstrates that a larger dipolar interaction can dominate the visible spectrum while the target altermagnetic source remains weak; MnTe and CrSb require broader exchange/symmetry sectors; hematite supplies a response-specific long-range exchange family; and bulk RuO₂ demonstrates that the framework must stop when the prerequisite material state is contradicted. MnSi supplies the complementary specificity limit: chirality itself is not a unique fingerprint of altermagnetism.

The hardened protocol is therefore

$$\begin{array}{l} \text{state/reference} \rightarrow \text{source} \rightarrow \text{probe} \\ \rightarrow \text{counterfactual} \rightarrow \text{specificity} \rightarrow \text{closure} \end{array} \quad (24)$$

Symmetry tells us what may exist. A predeclared adversarial benchmark tells us what has actually been identified.

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primary literature, explicit manuscript definitions, and rendered pages during preparation. The author retains full responsibility for all claims, equations, classifications, citations, figures, and conclusions.

DATA AVAILABILITY

No new experimental data were generated. All material-specific conclusions are based on the published or preprint

sources cited in the manuscript, with a literature cutoff of September 12, 2026.

COMPETING INTERESTS

The author declares no competing interests.

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